

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-VII (NEW) EXAMINATION – WINTER 2024****Subject Code:3174205****Date:11-12-2024****Subject Name: Natural Language Processing****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) State advantages and disadvantages of Natural Language Processing.	03
	(b) Describe the components of Natural Language Processing (NLP) and highlight their differences.	04
	(c) Explain phases of Natural Language Processing in detail along with an accompanying diagram.	07
Q.2	(a) Explain Add-one smoothing with example.	03
	(b) What is morpheme? Describe morphology parsing and its role within NLP.	04
	(c) Explain how Named Entity Recognition (NER) works in NLP, using the sentence: 'Apple Inc. was founded by Steve Jobs and Steve Wozniak in a garage'?	07
	OR	
	(c) Describe the N-gram technique and also explain Unigram, Bi-gram, and Tri-gram concepts using an example sentence: “learning is never done without errors and defeat”.	07
Q.3	(a) Explain what word embedding is, and list out different types of word embeddings and their subtypes.	03
	(b) Explain the Continuous Bag of Words (CBOW) model with an example.	04
	(c) Explain word sense disambiguation briefly.	07
	OR	
Q.3	(a) Explain the bag-of-words model and provide an example of it.	03
	(b) Explain skip-gram with the help of example.	04
	(c) Explain supervised word sense disambiguation briefly.	07
Q.4	(a) Explain sentiment mining and its type in brief.	03
	(b) Explain information retrieval.	04
	(c) Explain Text summarization and also explain different dimensions of text summarization.	07
	OR	
Q.4	(a) Provide a brief explanation of Question Answering in a multilingual setting.	03
	(b) Explain Cross-Lingual IR.	04
	(c) Explain information extraction in brief.	07
Q.5	(a) Discuss the necessity for machine translation.	03
	(b) Explain Neural Machine Translation.	04
	(c) Explain different machine translation approaches.	07
	OR	
Q.5	(a) Discuss the challenges of machine translation	03
	(b) What are the drawbacks of direct translation approach?	04
	(c) Explain encoder and decoder architecture for MT.	07
